OCEAN ENTERPRISES

Business Management System

Product, Operator & Technical Documentation

Live application	https://ocean-enterprises.vercel.app/
In-app documentation	https://ocean-enterprises.vercel.app/docs
Source repository	github.com/alicharolia7869/ocean-enterprises

Prepared from the repository documentation and application structure available on 29 October 2026. Confirm current production behavior before relying on operational or technical details.

1. About the System

Ocean Enterprises is a water supply and wholesale beverage distribution business operating in Mumbai. Its Business Management System is designed to bring daily water deliveries, beverage orders, dispatch, customer accounts, inventory and payment tracking into one web workspace.

The system supports high-frequency water-jar entries, multi-item beverage orders, customer ledgers, delivery status tracking, stock control and reporting. It is an operational aid; staff should review parsed orders and financial entries before saving or sending customer communications.

2. Main Modules

Module	Purpose
Dashboard	Operational KPIs, dispatch pipeline, active orders and low-stock alerts.
Rapid Water Entry	Record 20L jar deliveries, adjust quantities and rates, mark paid/pending, and view monthly customer summaries.
WhatsApp Orders	Paste a customer message into the rule-based parser, review extracted customer/products/quantities, then accept the order.
Deliveries & Route Map	Track dispatch stages, reorder stops, view mapped stops and open Google Maps navigation.
Customers	Maintain customer details and categories, capture GPS coordinates with browser permission, view ledgers and print statements.
Payments	Record full or partial payments using Cash, UPI, Bank Transfer or Cheque; general payments use FIFO allocation.
All Orders	Create and filter orders, update status, record payments, export CSV and cancel/delete orders.
Inventory & Stock	Manage water/beverage products, prices, stock quantities and low-stock thresholds.
Reports	Daily, weekly and monthly sales views; customer and product reports; CSV export and print/PDF.
Settings	Configure business profile details, UPI, contact information and administrator password.

3. Recommended Daily Workflow

1. **Check stock:** Review inventory and low-stock alerts before loading deliveries.
2. **Record water drops:** Use Rapid Water Entry and verify customer, jar count, rate and payment status.
3. **Process incoming orders:** Paste WhatsApp order text, review every extracted line item and confirm customer identity before accepting.
4. **Dispatch deliveries:** Use the Deliveries board to sequence stops and update order status as work progresses.
5. **Record collections:** Enter received payments and select the correct payment mode; check customer balances.
6. **Reconcile:** Review reports and export CSV or print customer statements as required.

4. Important Business Rules

• Review parsed WhatsApp orders before accepting them. The parser is rule-based and does not replace human verification.
• Payment entries can be partial. Confirm the amount and customer before saving.
• General payments are documented as using FIFO allocation: oldest unpaid orders first, then older unpaid water entries.
• Inventory behavior depends on the order or water-entry action; check stock after cancellation/deletion workflows.
• WhatsApp links open a pre-filled message for a user to review and send. This is not the same as automatic WhatsApp API delivery.

5. Technology & Architecture

Layer	Implementation documented in repository
Frontend	React 19, React DOM, Vite
Backend	Node.js and Express 5
Production database	PostgreSQL through the pg package; managed Neon connection configured through environment variables
Local fallback	JSON document-store adapter when a PostgreSQL connection string is not configured
Authentication	JWT bearer tokens and bcryptjs password hashing
Hosting	Vercel static frontend and serverless Express API
Testing	Node.js built-in test runner; frontend linting via Oxlint

6. Local Setup (Developer Guide)

Prerequisites: Git, Node.js compatible with the project (repository guide lists Node.js 20+), and npm. PostgreSQL is optional for local use if the JSON fallback is intended.

Clone the repository and install dependencies:

```
git clone https://github.com/alicharolia7869/ocean-enterprises.git
cd ocean-enterprises
npm install
npm --prefix client install
```

Create a local environment file from the example and edit it locally:

`copy .env.example .env` (Windows Command Prompt) or `cp .env.example .env` (macOS/Linux). Set the appropriate values privately. Never commit real secrets.

Common commands from the root directory:

`npm start` — start the backend using the project's server entry point.
`npm run dev:client` — start the Vite frontend.
`npm test` — run the Node.js test suite.
`npm run build` — build the frontend for deployment.
`npm run seed` — run the seed script; inspect its behavior and use only in an appropriate environment.

7. Security & Configuration

Environment variables are configured outside source control. The repository's .env.example documents variable names including PORT, NODE_ENV, JWT_SECRET and database connection settings. Do not copy real production values into documentation, screenshots or commits. Use a strong unique JWT secret in production and keep database credentials private.

Use production data carefully. Before migrations, recovery operations or destructive changes, ensure an appropriate backup exists and confirm the target environment. Never run seed/reset operations against production unless the intended effect is understood.

8. Troubleshooting Checklist

Cannot sign in: Verify the registered identifier and password; check server logs without exposing credentials.

Database connection fails: Confirm the database URL is configured in the intended environment and that the database is reachable.

Data does not persist: Confirm which storage mode is active. A local JSON fallback and managed PostgreSQL have different persistence behavior.

Build fails: Install dependencies, review the first meaningful error, and run the project's build command.

Map/GPS issue: Allow browser location permission and verify the customer has usable coordinates; external map navigation needs connectivity.

CSV/print issue: Check browser download/print permissions and retry with the relevant report or order filter.

9. Documentation Directory

The repository already includes a detailed Markdown documentation suite. Start with docs/index.md, then use the relevant guide:

- docs/business-overview.md — business context
- docs/features.md — feature inventory and limitations
- docs/user-guide.md — operator instructions
- docs/system-architecture.md — architecture
- docs/database-design.md — persistence and schema
- docs/api-reference.md — API routes
- docs/authentication-and-security.md — security
- docs/local-development.md — local setup
- docs/environment-configuration.md — environment variables
- docs/deployment-guide.md — Vercel deployment
- docs/testing-and-validation.md — test and lint procedures
- docs/troubleshooting.md — common issues
- docs/maintenance-and-recovery.md — maintenance
- docs/changelog.md — recorded changes

Source of truth: repository code and its Markdown documentation. This PDF is a convenient overview, not a substitute for checking current code, environment configuration, or production status.